

Aarya Venkat, PhD

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PROFESSIONAL SUMMARY

Computational chemist with over 2 years of industry experience in computational protein and drug discovery. Expert in integrating SOTA machine learning with physics-based modeling (MD, DFT, Docking) to accelerate protein-ligand discovery and optimization. Proven track record of building automated MLOps pipelines (Docker, Kubernetes) and leading cross-functional projects in fast-paced environments.

EDUCATION

University of Georgia — PhD Computational Chemistry — Dec 2023

Honors: ARCS Foundation Scholar, Future Faculty Fellow, Three Minute Thesis award

University of California, San Diego — MS Chemistry — June 2017

TECHNICAL SKILLS

Programming: Python, Bash, Git, Snowflake, Docker, Kubernetes, Argo Workflows, Nextflow, AWS/GCP

Bioinformatics: Metagenomic Sourcing, Sequence analysis, Intron/Exon analysis, NGS, Clustering

Cheminformatics: RDKit, QSAR, xTB, CREST, molecular docking, virtual screening

Computational Chemistry: Molecular Dynamics, DFT, MDAnalysis, Amber, GROMACS, Desmond, PyMol, Schrodinger Maestro, forcefield parameterization, BioEmu, protein-ligand interaction analysis

Protein Design: Inverse Folding (ProteinMPNN, LigandMPNN), Mutational inference (ThermoMPNN, ESM2), co-folding, de novo protein design (RFDiffusion), training ML models for enzyme optimization

Deep Learning: PyTorch, Explainable AI (SHAP), linear/logistic regression, Protein Language Models (ESM2), fine-tuning (PEFT, LoRA), Predictive modeling, Dataset preparation

Research & Soft Skills: Technical Writing, Data Visualization and Presentation, Public Speaking

PROFESSIONAL EXPERIENCE

Ginkgo Bioworks, Inc. — Scientist III

Dec 2023 – Present — Boston, MA

- Lead over 30 protein engineering campaigns utilizing molecular docking, modeling, and evolutionary models (ESM2, EVCouplings) to enhance specificity, thermostability, and activity for high-budget contracts.
- Serve as Subject Matter Expert and Tech Liaison for AI and computational biology, leading customer-facing meetings to communicate research benefits, risks, and strategic roadmaps.
- Developed **PyTorch** deep learning models fine-tuned via **LoRA** from ESM2 to build family-specific enzyme models for Ginkgo's enzyme custom plates service.
- Architected **Docker** and **Kubernetes** pipelines for **RFDiffusion-AA**, proteinMPNN, and proprietary large-scale protein design tools.
- Utilized metagenomic sourcing and structure-prediction tools (ESMFold, AlphaFold, Boltz) to achieve protein function objectives.

University of Georgia — Graduate Research Assistant

July 2018 – Dec 2023 — Athens, GA

- Conducted advanced evolutionary analyses including molecular dynamics, phylogeny, and quantum chemistry to study enzyme function.
- Developed deep learning models and tools to classify enzyme function.

- Authored 17+ peer-reviewed manuscripts, contributing to writing, experimental design, data visualization, and successful funding acquisitions.
- Managed lab operations like data storage infrastructure, procurement, and protocol documentation.

AWARDS

2023 — Eriksson Lecture Scholarship
 2023 — ASBMB Conference Travel Award
 2022 — Glycobiology Poster Award
 2022 — Glycobiology Travel Award
 2021 — Glycobiology Poster Award
 2021 — Glycobiology Travel Award
 2021 — UGA Travel Award
 2021 — ARCS Foundation Scholar
 2021 — Three Minute Thesis Award

SELECTED PUBLICATIONS

Total: 22 — Citations: 409 — H-index: 11 — co-first author[†]

Venkat, A. (2026). GapClean, a tool for cleaning and improving comprehension of protein sequence alignments. *In Prep.*

Fang, R. et al. (2026). Glydentify: An explainable deep learning platform for glycosyltransferase donor substrate prediction. *Submitted.*

Zhang, J. et al. (2025). ICEPIC: A Toolkit to Discover Ice Binding Proteins from Sequence. *bioRxiv.*

Venkat, A. et al. (2023). Mechanistic and evolutionary insights into isoform-specific ‘supercharging’ in DCLK family kinases. *eLife.*

Zhou, Z. et al. (2023). xAI meets biology: A comprehensive review of explainable ai in bioinformatics applications. *arXiv.*

Meng, Y. et al. (2023). Phosphorylation-dependent pseudokinase domain dimerization drives full-length MLKL oligomerization. *Nature Communications.*

Aceil, J.[†], **Venkat, A.**[†], et al. (2023). Prevalence and homology of the pneumococcal serine-rich repeat protein at the global scale. *Microbiology Spectrum.*

Venkat, A., et al., (2022). Modularity of the hydrophobic core and evolution of functional diversity in fold A glycosyltransferases. *Journal of Biological Chemistry.*

Taujale, R., **Venkat, A.**, et al. (2020) Deep evolutionary analysis reveals the design principles of fold A glycosyltransferases. *eLife.*

PATENTS

Venkat, A., Castro, Y., and Manning, T. Using Computational QSAR Methods to Propose a New Group of Antibiotics for Dental Applications, U.S. Provisional Patent Appl. 62/359,638, July 22, 2016.